Second-Order Conditions, Concavity, and Convexity

ECON 441: Introduction to Mathematical Economics
Lecture 15

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Recap

For a continuously differentiable function $y = f(x)$:

- A critical point x_0 satisfies $f'(x_0) = 0$ (necessary for an interior extremum)
- First-derivative test: check whether $f'(x)$ changes sign at x_0

Today: a quicker test that uses the **second derivative**, and functions for which a critical point is guaranteed to be a global optimum.

Second and Higher Derivatives

The derivative of $f'(x)$ is called the **second derivative**:

$$f''(x) = \frac{d^2y}{dx^2}$$

Similarly, we can obtain higher-order derivatives:

$$f'''(x),\ f^{(4)}(x),\ \ldots,\ f^{(n)}(x) \qquad \text{or} \qquad \frac{d^3y}{dx^3},\ \frac{d^4y}{dx^4},\ \ldots,\ \frac{d^ny}{dx^n}$$

Example: find $f''(x)$ for $f(x) = x^3 - 12x^2 + 36x + 8$

What the Second Derivative Tells Us

With an infinitesimal increase in x from x_0:

- The **value** of the function increases if $f'(x_0) > 0$ and decreases if $f'(x_0) < 0$
- The **slope** of the function increases if $f''(x_0) > 0$ and decreases if $f''(x_0) < 0$

Second-Derivative Test

If $f'(x_0) = 0$, then $f(x_0)$ is

1. a **maximum** if $f''(x_0) < 0$
2. a **minimum** if $f''(x_0) > 0$

More convenient to use than the first-derivative test.

What if $f''(x_0) = 0$?

Example

On Monday we found the critical points $x = 2$ and $x = 6$ of

$$f(x) = x^3 - 12x^2 + 36x + 8$$

Classify them using the second-derivative test.

Example

$$f(x) = x^4 - 4x^3$$

Find the critical points. What does the second-derivative test say at each one?

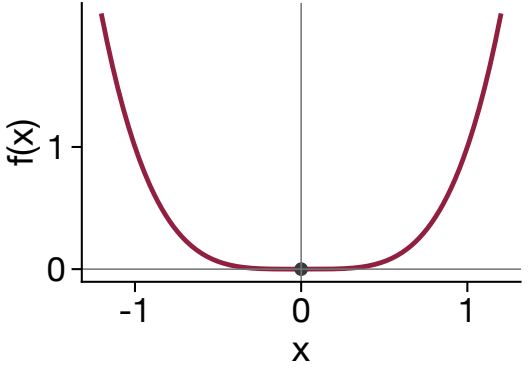
Necessary vs Sufficient Conditions

For a **maximum** at x_0:

- First-order necessary: $f'(x_0) = 0$
- Second-order necessary: $f''(x_0) \leq 0$
- Second-order sufficient: $f''(x_0) < 0$

For a **minimum**, reverse the inequalities. The second-order conditions apply only after the first-order condition is satisfied.

$y = x^4$



Concave and Convex Functions

- **Concave** function: $f''(x) \leq 0$ for all x
- **Convex** function: $f''(x) \geq 0$ for all x
- **Strictly concave** function: $f''(x) < 0$ for all x
- **Strictly convex** function: $f''(x) > 0$ for all x

Concave and Convex Functions

f is **concave** if, for any x_1, x_2 and $\lambda \in (0, 1)$,

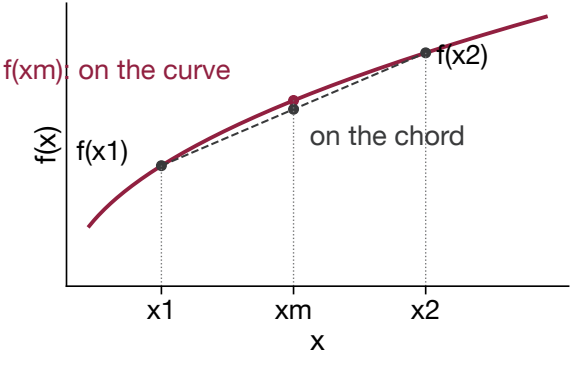
$$f(\lambda x_1 + (1 - \lambda)x_2) \geq \lambda f(x_1) + (1 - \lambda)f(x_2)$$

f is **convex** if

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$$

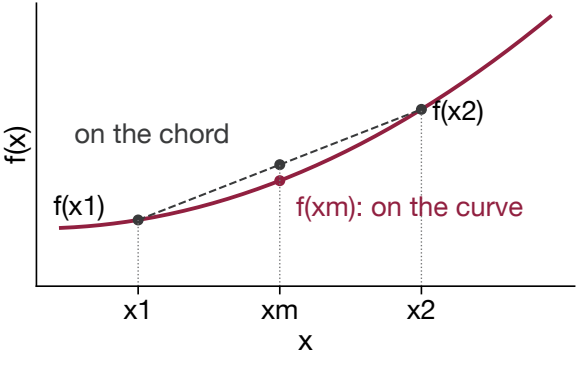
For strict concavity or convexity, replace with strict inequalities.

Concave Function



At $x_m = \lambda x_1 + (1-\lambda)x_2$, the curve lies **above** the chord:
$f(x_m) \geq \lambda f(x_1) + (1-\lambda) f(x_2)$

Convex Function



At $x_m = \lambda x_1 + (1-\lambda)x_2$, the curve lies **below** the chord:
$f(x_m) \leq \lambda f(x_1) + (1-\lambda) f(x_2)$

Attitudes toward Risk

Consider the following game: flip a coin, collect $0 if tails, collect $20 if heads.

How much would you pay to play this game?

If your utility from money $u(m)$ is concave, how does $u(10)$ compare with $\frac{1}{2}u(0) + \frac{1}{2}u(20)$?

Example

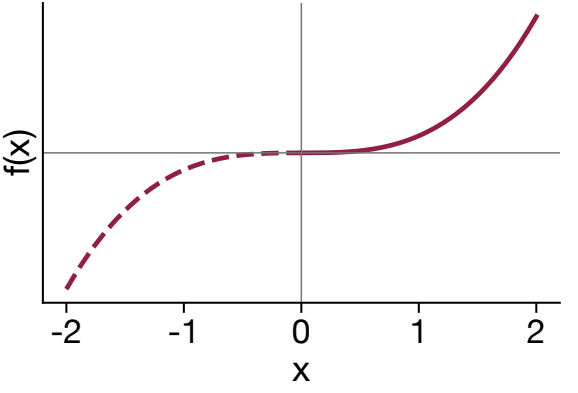
The domain of f is all real numbers:

$$f(x) = x^3$$

Is f convex, strictly convex, concave, or strictly concave?

What if the domain of f is all nonnegative real numbers?

$f(x) = x^3$



Properties of Concave and Convex Functions

1. A linear function is both concave and convex, but not strictly so.
2. If $f(x)$ is (strictly) concave, then $-f(x)$ is (strictly) convex, and vice versa.
3. If $f(x)$ and $g(x)$ are both concave (convex), then $f(x) + g(x)$ is concave (convex). If, in addition, one or both of them are strictly concave (convex), then $f(x) + g(x)$ is strictly concave (convex).

Global Optimizers

- If a function is concave, any critical point gives a global maximum.
- If a function is strictly concave, any critical point gives the **unique** global maximum.
- If a function is convex, any critical point gives a global minimum.
- If a function is strictly convex, any critical point gives the **unique** global minimum.

Example

Say $f(x)$ is a strictly concave function and

$$f'(2) = 0$$

Is $f(2)$ a local or a global maximum? Is it the unique maximum?

Example: Profit Maximization

A firm's profit is $\pi(Q) = R(Q) - C(Q)$ with revenue $R(Q) = 20Q - Q^2$ and cost $C(Q) = 4Q + 10$.

Find the profit-maximizing output. Is it a global maximum? Why?

Homework Questions

- Exercise 9.3: 2, 3, 4, 5
- Exercise 9.4: 1 (b), (d), 2, 3, 5

Textbook reference: 9.3, 9.4